GridStemLayNet - A Document Layout Understanding Model

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***Abstract*—Document Layout Understanding and Analysis solves the real-time problem of observing and contextualizing Layout and textual characteristics of a document. Document pre-trained models have proven to be very effective on various tasks in Document AI. However, for the document layout analysis (DLA) task, existing document pre-trained models, even those pretrained in a multi-modal fashion, usually are large and rely on complex visual features and just try to segment or contextualize high-level layouts but does not capture low level textual features. This research proposes a model that encompasses both layout features and textual features. To fully leverage multi-modal information, it uses Global Invariance and exploits efficient training techniques to learn better representation for DLA and textual properties. This research proposes a vision transformer based approach with Stem Attention mechanism. For the purpose of this research a new dataset called DTFiL will also be introduced for this specific problem.**

I. Introduction

Over the past few years, substantial progress has been made in the field of Document AI with the development of document pre-trained models [15] and grid-based models [6, 26], which have proven to be effective in addressing a wide array of tasks. From simple document understanding to complex information extraction tasks, these models have exhibited considerable generalization capabilities. However, such pre-trained models of documents are likely to fail in capturing low-level textual characteristics with multi-modal approaches for document layout analysis (DLA).

This research focuses on incorporating textual characteristics as part of the Layout Analysis process, the general layout analysis process involves predicting blocks of texts and charts and images, what this research tries to introduce is textual characteristics prediction as part of the Layout Analysis process, textual characteristics refers to a set of properties associated with a text i.e. whether the text is Bold, Italic or Underlined. The research fosters a more profound comprehension of document layout analysis, emphasizing visual and textual properties in a unified framework through the synergy of multimodal information.

The motive for this research is to drive the research into DLA forward and expand the scope for improvement by introducing a new problem space i.e. textual characteristic understanding, for the purpose of this research a new dataset is introduced, DTFiL. To address the problem, multiple experimentations were conducted and a set of architectures were trained. The models were trained on a wide variety of datasets, including FUNSD [14], PubLayNET [28], and custom DTFiL dataset, guaranteeing a solid and versatile foundation to capture the most intricate details of document layout in different domains.

The final architecture is explained in this paper, which unifies convolution-grid-based approach and a Vision Transformer approach for understanding textual features. In doing so, it can capture the benefits of both vision transformers and traditional convolutional and grid-based models. This holistic framework allows the model to adjust very well to the different structures coming from different layouts of the documents.

The rationale behind the chosen vision transformer-based approach, detailing the training methodology and datasets is articulated. Additionally, exploration of convolution and grid-based techniques for textual feature understanding are introduced. This work, aspires to contribute to the advancement of document AI by pushing the boundaries of multi-modal analysis in the realm of document layout understanding.

II. Related Works

For a very long time, document layout analysis, or DLA, has been an important area of study in computer vision. Rule-based approaches were predominant at first, using basic

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properties like texture or picture pixels directly for layout analysis and comprehension. Low-level visual cues were used by later machine learning techniques [7, 9] to parse page layout. Convolutional neural networks [11] and other vision-based modeling techniques have been used to address DLA as a visual object recognition or segmentation problem more recently with the advent of deep learning [18, 12, 17, 22].Document AI has evolved greatly with the advent of self-supervised pre-training methodologies [24, 25, 2, 19, 13, 15, 10, 20]. With notable success, a number of document pre-trained models [15, 10, 13] have been used for DLA. For instance, DiT [15], which was motivated by BEiT [4], trains a document image transformer for layout analysis and demonstrates encouraging outcomes, but it ignores the textual content of documents. On the other hand, documents are modeled by UniDoc [10] and LayoutLMv3 [13] utilizing a single architecture that combines text, vision, and layout modalities. But they do not include text embeddings for object detection while fine-tuning DLA; instead, they rely solely on the vision backbone. Unlike vision-centric approaches, LayoutLM [25] treats DLA as a sequence labeling task and focuses only on the text modality. Results from the LayoutLM experiments revealed that NLP based methods can work for DLA. Nevertheless, these techniques are often extremely unimodal or even solely unimodal because they emphasize visual information at the expense of textual input. These methods generate text grids (text embedding maps) using text and layout data and combine them with visual elements. For the DLA challenge, Yang et al. [26] created a sentence-level grid that is paired with visual cues. Two-stream CNNs are used in VSR [27], where the visual stream and semantic stream, respectively, process text grids (char-level) and images (sentence-level). But these approaches only use text grids as extra features or model inputs; they don't provide semantic supervision for DLA training, which makes it difficult to achieve meaningful semantic comprehension. But none of the previous works discussed above include textual characteristics modeling, This research focuses on integrating textual level visual characteristic comprehension into the DLA process.

In the past, datasets for DLA [26, 1, 5] were rather small and mostly comprised of non-technical or technical articles from news dailies, weeklies, or technical journals. The development of large-scale datasets like DocBank [16] and PubLayNet [28] has brought great progress to DLA research. Whereas PubLayNet [28] has 360K scientific papers with 5 layout types, including text, title, list, figure, and table, DocBank [16] contains 500K scientific paper documents with 12 types of layout units. These databases are primarily composed of scientific publications, which restricts the range of document kinds and layout categories even with their contributions. Furthermore, these datasets' arrangement categories—which are intended for scientific papers—present difficulties when used to other kinds of documents in practical settings.

III. Existing Data Sets

A. AdobeVFR

The AdobeVFR Dataset comprises of various fonts for font analysis and recognition tasks in the field of graphics design and document processing. It has a wide range of about 2,383 classes of popular fonts. This offers a diverse range of typographical styles commonly encountered in various design contexts.

One notable feature of this dataset is that it includes synthetic training data. These synthetic text images are tightly cropped, so that the focus remains solely on the font itself without distracting elements. Additionally, the images are converted to grey-scale and size-normalized, which helps fabrication of the training data and makes it suitable for training machine learning models. This dataset is proved to be helpful for researchers and practitioners which helps them to develop and evaluate font recognition algorithms, font matching systems, and other applications related to typography and graphic design. Its vastness makes it a very useful resource for the development of state-of-the-art in font analysis and recognition technology.

Problem: It does not contain the entire document image, instead, the image of the text is cropped and provided.

B. FUNSD

The FUNSD [14] dataset comprises of a rich collection of documents for training and evaluating document understanding systems. It has various document formats, including business invoices, bill receipts, and forms, Each document in the dataset is annotated, providing crucial information such as text blocks, fields, and their interrelations. These annotations serve as ground truth data for training machine learning models, to accurately recognize and extract relevant information from the input documents. This dataset can be used for training and development in Document Layout Analysis.

Problem: It only focuses on the layout of the documents related to forms. It not only lacks the other document types but also is small in size.

C. PubLayNet

The PubLayNet [28] dataset is a compiled resource of data for document layout analysis, developed by the Allen Institute for Artificial Intelligence. It is vast in scale, comprising over one million document images with detailed layout annotations. These annotations offer detailed information about the structure and arrangement of all the elements within each document image. This dataset includes images of research papers, patents, newspaper articles, and various other document formats commonly used by various sectors. The primary objective of this dataset is to provide a comprehensive data for training and evaluating algorithms pertaining to document analysis tasks. PubLayNet is helpful with the advancement of machine learning and computer vision algorithms in the domain of document analysis as it offers a rich and diverse set of annotated document images.

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Problem: This dataset only contains images that are specifically inclined towards DLA. This doesn't have anything about Font characteristics.

IV. Our Dataset

A. Methodology

The main concept of This dataset creation process involves an exhaustive gathering of PDF documents from various resources and domains. Various repositories and databases were explored to obtain a compiled and varied assortment of PDFs, spanning from academic journals to technical manuals, corporate reports, and more. This compilation step ensures a broad representation of textual data, crucial for an efficient model training. Upon acquisition of the required data, each PDF is pre-processed, by organizing and structuring the documents within a repository. This preparatory step is helpful with the optimization of the PDFs for subsequent processing and ensuring uninterrupted extraction of relevant textual information while preserving the document's integrity.

With the help of PyMuPDF library, the structure of each PDF document is analysed with its contents parsed, to extract the textual elements and their corresponding bounding boxes. This parsing phase involves a thorough analysis of the document's layout, with the observation of individual text entities and their spatial arrangements within the pages. As the parsing progresses the textual content is extracted with crucial metadata related to font characteristics. This includes acquiring necessary information of the text, such as typeface, size, style, and weight, which is vital for subsequent modifications. This process of cataloguing font attributes ensures fidelity of textual representation across diverse documents. The variability is introduces through systematic font modification in addition to the extracted textual content and font characteristics.

With a randomized augmentation approach, the font attributes within each bounding box are manipulated programmatically to induce diverse variations. This strategy encompasses alterations in typeface, size, weight, and style, enriching the dataset with a wide spectrum of textual nuances. A block-level font variation strategy is used to maintain coherence within the document. This approach enhances dataset richness and diversity without compromising on coherence. After font manipulation, the PDF documents are reconstructed as grayscale images, each page representing a unique variation of the original content. With seamless integration of modified text into the image format, each labelled image encapsulates a distinct rendition of the underlying document and is enriched with diverse font characteristics.

B. Specifications

The dataset generation procedure involves a subset of 256 PDFs, each averaging 8 pages. These documents represent various content types including resumes, articles, reports, and invoices. Each PDF undergoes a series of 10 font randomizations per page, resulting in the creation of approximately 16,400 labelled images. This approach enhances dataset uniqueness and richness which also fosters model robustness by introducing ample variability.

The scalability of this methodology is profound, as it allows for the potential expansion of the dataset size. However, the max limit of scalability is determined by some factorial constraints such as computational resources and the availability of diverse PDFs from disparate domains.

V. Model Experimentation

In the pursuit of advancing DLA within the domain of document AI, this research has delved into a series of experimental model architectures. The overarching objective is to explore diverse strategies that can effectively capture both textual and visual characteristics inherent in document layouts. The challenges associated with DLA demand a nuanced understanding of layout modeling, class modeling, and the intricate interplay between these elements.

A. Experimentation 1: Convolution-Based Model

In the initial experimentation, a convolution- based model tailored for the task of DLA and textual characteristic understanding was developed. This model was specifically trained on textual characteristics modeling, leveraging the inductive biases inherent in general convolutional neural networks. Key to this approach was the utilization of the PubLayNET [28] dataset, which provided a rich source of diverse document layouts for training. The model demonstrated commendable performance, producing promising results that underscore the efficacy of convolution-based techniques in capturing textual features within document layouts.

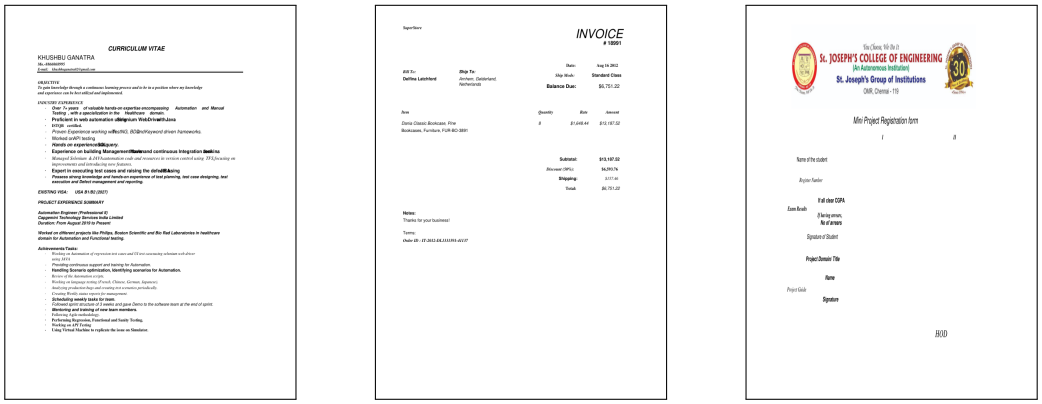
The simple well traversed architecture i.e. an encoder-decoder architecture with attention, where the encoder is a series of contextualizing and down-sampling layers characterized by convolution blocks as in the U-Net architecture [23], but with GeLU non-linear activations, and followed by a decoder block with linear layers and 2 different heads for different modeling i.e. block class modeling head, bounding box regression head.

$$\text{reghd, blkclshd} = Dec(Concat(Enc(X), Res52(X)) \quad (1)$$

Where X denotes the input data which will be a document image, $Enc(X)$ is the encoding function, $Concat(Enc(X), Res52(X))$ represents the concat mechanism applied to the encoded data and the feature extracted from the pretrained resnet architecture, and $Dec(Concat(Enc(X), Res52(X)))$ is the decoding process with multiple heads.

Experimentation 2: Vision Transformer-Based Approach Building upon the success of the initial convolution-based model, this experimentation ventures into the realm of vision transformers. This approach harnessed the principles of general transformers and attention mechanisms, successfully applying them to the challenges posed by DLA. Concentrating on both layout modeling and class modeling, the model was

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(a) (b) (c)

Fig. 1: Visually-rich documents with different layouts and formats from DFTiL, for training GridStemLayNet

trained on the DTFiL dataset. While the model performed considerably in class modeling, the model did not perform as well as it was expected to, in layout modeling. The architecture of the model is based on the paper of Alexey Dosovitskiy et al [8], experimentation was done with a swin transformer-based approach but both approaches led to some serious challenges to be encountered in the precise modeling of bounding boxes and detection of the class, especially in the swin-transformer approach as the patching hyper-parameters had significant impact on the translation invariance. The poor performance of this model could not be effectively determined, because of the size and complexity of Vision Transformer based models.

The model was modeled using SmoothL1RegLoss introduced in for bounding box modeling and general Cross-Entropy loss for class modeling provided insights into the nuanced aspects of DLA.

$$\text{ViT}(X) = Dec(Mha(Enc(X))) \tag{2}$$

$$\text{Swin}(X) = Ln(Swin(Mha(PatchMerging(Ln(Enc(X)))))) \tag{3}$$

(2) Where X represents the input image patches, $\text{Enc}(X)$ is the encoder process, MHA denotes the multi-head self-attention mechanism, and Dec represents the decoder part.

(3) Where X represents the input image, $\text{Enc}(X)$ is the encoder process, LN denotes layer normalization, PatchMerging represents the patch merging operation, and MHA denotes the multi-head self-attention mechanism.

B. Experimentation 3: Grid-Based Modeling Technique

This model is based on the assumption that attention mechanism will be able to capture complex intricate features learned from a convolution based encoder and ViT Encoder [8] and then combing them to model multiple outputs, the layout (or block) bounding boxes, class of the layout modules(whether it is an image or text block(paragraph) or table) followed by the text level class modeling, it uses a grid-based Modeling technique for the input images introduced in the paper GiT, a novel approach where it extracts bounding box features of each text present in an image document and represent them as a model input in the form of a grid-image tensor. This method allows establishes a baseline model that focuses on both layout modeling and class modeling and low-level textual features. By representing textual features document in a grid format, it aims to capture the spatial relationships and contextual information inherent in document layouts, providing a unique perspective on DLA.

In this model the concept of stem-attention is introduced which is a combination of multiplicative attention and scaled dot product attention, the idea is to harness the power of attention based networks. In the architecture, the ViT encoder and the grid encoder (convolution based encoder) both return a $\{(16, d_k)\}$ feature vector, where the d_k is set to 512 as a standard scheme.

First multiplicative-attention (3) is performed on the grid encoder output and then scaled dot-product based attention (5) is performed between the the output of the multiplicative attention operation and the the ViT encoder output and then the softmax probabilities is scaled on the additive result of both ViT encoder and grid encoder, the multiplicative-attention here has learnable parameters which helps in improving the performance. A series of Layer normalization and linear layer with GeLU activation are introduced to normalize and control the gradient flow, which in turn helps with the learning process and then it has three modeling heads.

$$\text{Matt}(h_i, s_j) = h_i^\top W s_j = K \tag{4}$$

$$\text{V} = K + V_o \tag{5}$$

$$\text{Attention}(V_o, K, V) = \text{softmax}\left(\frac{V_o K^\top}{\sqrt{d_k}}\right) V \tag{6}$$

Where in (4) h_i and s_j are two different linear projections of the grid encoder output (G_o), and in (5) V represents the additive result of G_o and V_o, and in (6) V_o is the ViT encoder output and d_k is the hidden dimension.

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As this experimental journey unfolds, each approach contributes to the broader understanding of the challenges and nuances in DLA. By combining insights from convolution-based models, vision transformers, and grid-based techniques, this research strives to develop a comprehensive model that excels in capturing both textual and visual features within diverse document layouts. These experimentations pave the way for future advancements in the field of document AI, addressing the complexities of DLA with innovative and effective model architectures.

VI. Architecture of Our Proposal

This model uses the vision transformer as the base of the architecture and builds upon it. The model has two encoder block, the vision transformer based encoder and grid encoder, the former is same as in the paper by Alexey Dosovitskiy et al [8] and the latter is series of double convolution block and max-pool down-sampling as is Fig 2, the over all architecture of the model is shown in Fig 4.

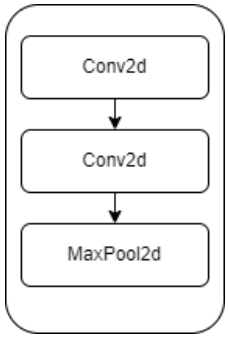


Fig. 2: Encoder Block of the grid encoder

Another core component of the architecture is the stem-attention block shown in Fig 3. and it works as explained before, it first performs multiplicative-attention on the grid encoder output and then perform scaled dot-product based attention between the the output of the multiplicative attention operation (the multiplicative attention here has learnable parameters as mentioned earlier this allows the model to learn significant features between grid patterns) and the ViT encoder output and then scale the softmax probabilities on the additive result of both ViT encoder and grid encoder, then it uses a series of normalization and linear layer and then it has three modeling heads.

The normalization layers are Layer Normalization[3], which plays a crucial role by normalizing across layers, as the weights from the previous layers can follow any kind of distribution it is paramount to scale and center them to follow a Gaussian distribution, so that the upcoming activations don't get saturated along the flat profile leading to dead gradients.

The stem attention block of the architecture is followed by a feed forward network, this feed forward network helps to learn the final representation of the image and

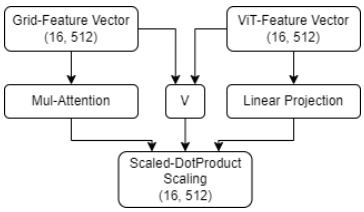


Fig. 3: Encoder Block of the grid encoder

layout and textual features, the feed forward network is composed of Linear Projections, GeLU activations and Layer Normalization. This feed forward network is followed by 3 prediction heads, each head is projected using a series (2) of feed forward networks these layers are followed by a leaky relu activation which allows a small negative slope and introduces a level of non linearity in the network and helps to capture and be able to understand more complex features, the final modeling is characterized by one regression head BMH (Bounding Box Modeling Head) and two classification heads ChMH and ClsMH(text Characteristics modeling and Layout class Modeling), the classification head is penalized by CrossEntropy Loss and the regression head by SmoothL1Loss. the choice for SmoothL1Loss is reinforced by the fact that it is less sensitive to outliers than MSELoss and in some cases prevents exploding gradients, the SmoothL1Loss can be represented as follows.

$$\ell(x, y) = L = \{l_1, \ldots, l_N\}^\top \tag{7}$$

$$\mathrm{l_n} = \begin{cases} 0.5(x_n - y_n)^2/\text{beta}, & \text{if } |x_n - y_n| < \text{beta} \\ |x_n - y_n| - 0.5 \times \text{beta}, & \text{otherwise} \end{cases} \tag{8}$$

`SmoothL1Loss` can be viewed as `L1Loss`, but with the $|x-y| < \beta$ portion replaced by a quadratic function such that its slope is 1 at $|x-y| = \beta$. The quadratic segment smooths the `L1` loss near $|x-y| = 0$ [21].

In essence, this research seeks to propel document AI towards a future where advanced models seamlessly integrate various aspects of analysis, providing a unified solution for tasks ranging from layout comprehension to interactive information retrieval.

VII. Results

The journey through diverse model experimentation for DLA underscores the multifaceted nature of this complex task. When it comes to the first experimentation the convolution-based modeling approach, the model results emphasized more on higher layout level performance which was recorded in good block modeling score, but when it comes to the convolution-based model there was a serious shortcoming in the characteristic modeling scores and even though it did well in block modeling score, it again failed to do well in the class modeling score which is separate for each detected blocks, this more or less ended up as the limitation for the convolution

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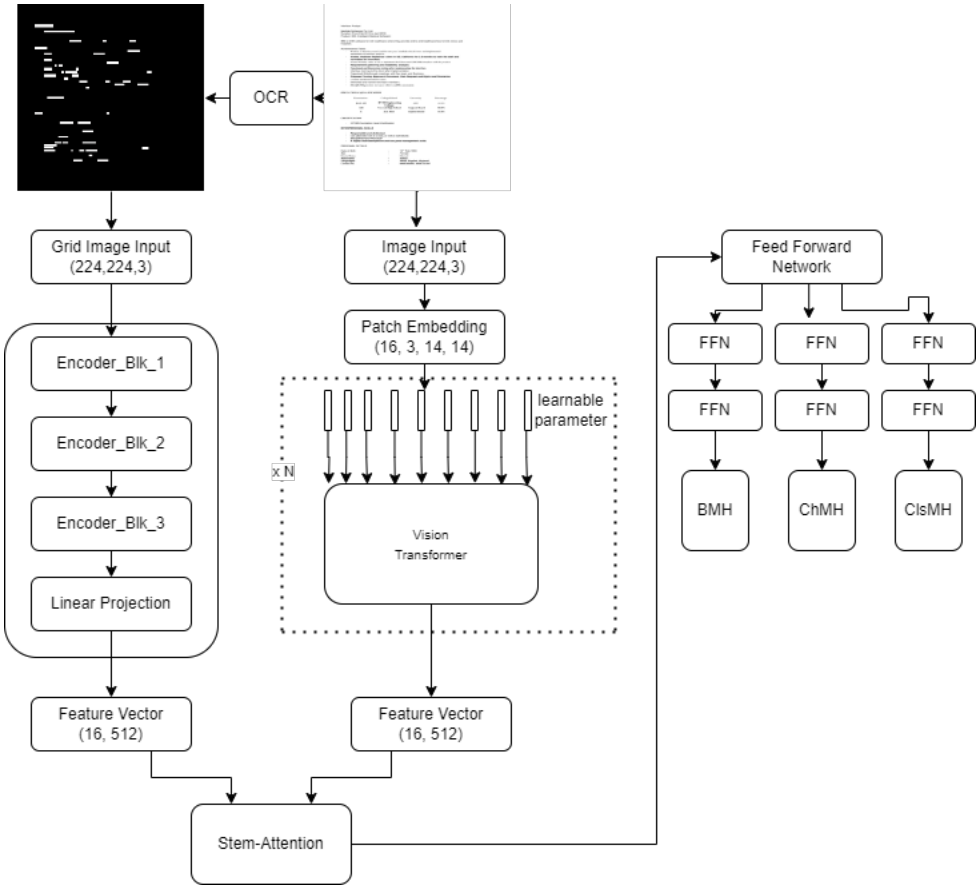


Fig. 4: Architecture of the model

based approach, Fig 5. shows the training performance of the model over 30 epochs.

The second experimentation involves using a straight forward vision transformer, this approach was also able to encode high level features such as the individual block layout and also performed really well in class modeling, again this model approach also failed in capturing low level textual features which is one of the most important part of this research, you can see the performance of the models in these plots, both the first experimentation referring to the CNN approach and the second experimentation referring to the vision transformer(Fig 5., Fig 6.), did not do really well in class modeling.

TABLE I: Experimentation scores on different datasets.

Model	DFTiL	FUNSD	PubLayNet
Experiment 1	30.3	34.22	32.03
Experiment 2	35.23	24.46	45.4
Experiment 3	65.52	34.43	62.52

Table I shows the performance of various experimentation along different datasets, namely the DFTiL, FUNSD and PubLayNet.

TABLE II: Experimentation scores on F1, Recall and Precision.

Model	Precision	Recall	F1-Score
Experiment 1	52.34	37.21	43.5
Experiment 2	51.67	22.20	30.91
Experiment 3	53.33	57.28	55.23

Table II shows the performance of various experimentation along different metrics, namely the F1 score, recall and precision as average of class and characteristics prediction heads.

Table III shows the performance of various established models on the IOU score, the IOU score here references the overlapping between the predicted block bounding box and the ground truth, these scores shows that the model is not on-par with the other models, since it was trained on a much smaller dataset, the major success is that this research shows that even low level textual characteristics can be learned in a supervised fashion along with layout features, the F1 score for the textual characteristics prediction shows that the model has

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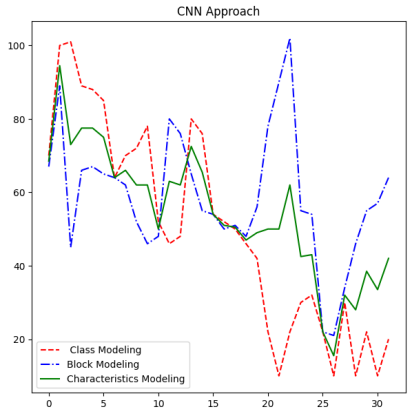


Fig. 5: Training Performance of Experimentation 1

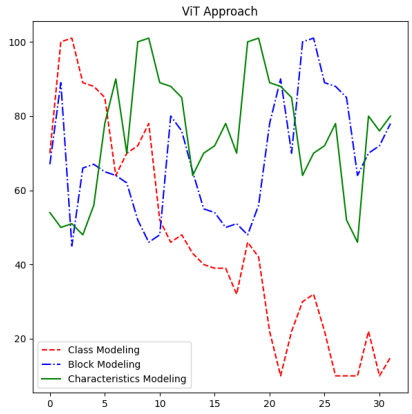


Fig. 6: Training Performance of Experimentation 2

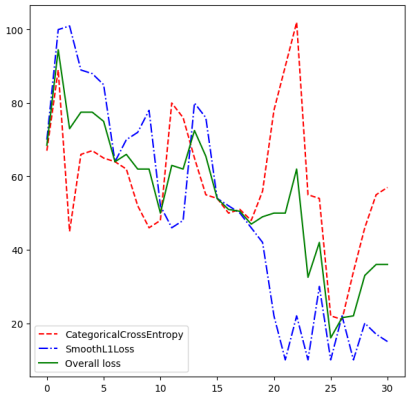


Fig. 7: Training Performance of Experimentation 3

TABLE III: DLA mAP @ IOU on PubLayNet validation set. GridStemLayNet is shortened to GSLayNet, Textual Characteristics with F1 score for the Publaynet validation set.

Model	Paragraph	Title	Lists	Figure	Table	Textual Charcteristics
ResNext	0.916	0.845	0.918	0.971	0.952	-
DeiT-B	0.934	0.874	0.921	0.972	0.957	-
BEiT-B	0.934	0.866	0.924	0.973	0.957	-
MAE-B	0.933	0.865	0.918	0.973	0.959	-
DiT-B	0.934	0.871	0.929	0.973	0.967	-
DiT-L	0.937	0.879	0.945	0.974	0.968	-
GSLayNet	**0.832**	**0.677**	**-**	**0.774**	**0.892**	**0.734**

set a benchmark for what it was trained for, On the contrary as the model did not perform well on data out side the training distribution especially the FUNSD dataset.

The exploration of grid-based techniques for spatial understanding as well as low level feature understanding, combining the capabilities of both a vision transformer with its global in-variance capability and CNN based model's capability to capture low level features (in this case text characteristics) and arriving at an overall good performance across the board. The model is good in reducing the loss corresponding to the characteristics modeling head, and also performed good in both block modeling and class modeling heads, Fig 7. shows the performance of the grid-based approach before.

The predictions from the third experimentation shows a respectable performance than the other experimentations, Fig 8. and Fig 9. shows the best predictions of the models where the images are from the same distribution as the dataset used to train the model. The model also presented huge challenges when it comes to data from distributions outside the training distribution. Fig 10. and Fig 11. shows the limitations of this model, as the model has predicted the same set of bounding boxes and the classification head predicted flag characters (here flag characters are simply empty characters given by a single space), this behaviour points to the poor performance when it comes to generalization, this serious shortcoming in performance is being addressed in future works, The other major limitation is the poor interpretability of the model as it hinders informed improvements, the final significant limitation is for some test documents, the predictions where the exact same as in Fig. 9. and Fig. 10. this could not be addressed as the documents where from a different distribution than the training distribution coupled with the poor interpretability, this will also be addressed in future works, The main focus of this research is to introduce the problem of capturing textual characteristics along with Layout modeling, which has shown a significantly good performance.

VIII. Conclusion

In conclusion, the journey through diverse model experimentation for DLA underscores the multifaceted nature of this complex task. From convolution-based models emphasizing textual characteristics to vision transformers excelling in class modeling, and the exploration of grid-based techniques for spatial understanding, each experimentation contributes valuable insights to the broader field of document AI.

As there is significant room for improvement in the model's performance, in the upcoming works short comings in the dataset and the model performance will be addressed, The envisioned further advancements represent the next frontier in the research. The pursuit of a comprehensive multi-model architecture and MIM(Masked Image Modelling) [4] based self-supervised training will be the foundation for upcoming works and commitment to address the evolving needs of document analysis.

In essence, this research seeks to propel document AI towards a future where advanced models seamlessly integrate various aspects of analysis, providing a unified solution for tasks ranging from layout comprehension to low level textual characteristics understanding. Through these endeavors, this research contributes to the ongoing evolution of document

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AI, pushing the boundaries of what is achievable and paving the way for more sophisticated, user-friendly, and powerful document analysis systems

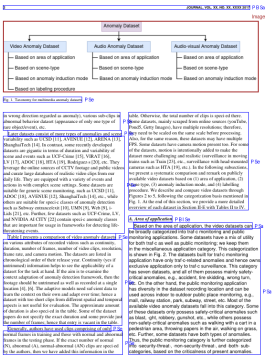


Fig. 8: Model prediction 1.

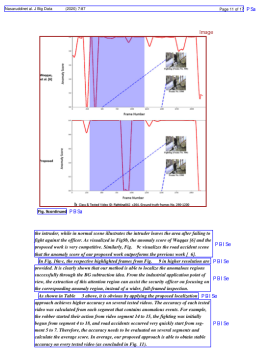


Fig. 9: Model Prediction 2.



Fig. 10: Model prediction 3.



Fig. 11: Model Prediction 4.

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